ACTIN Report (research use only)

PATIENT
EXAMPLE-LUNG-01

REPORT DATE
17-Sep-2025

Gender: **Female** | Birth year: **1975** | WHO: **1**

Tumor: **Lung adenocarcinoma** | Lesions: **Liver, Lung** | Stage: **IV**

Clinical summary

Relevant systemic treatment history	**2023-06 to 2025-01**	**Osimertinib**
Relevant other oncological history	**None**	
Previous primary tumor	**None**	
Relevant non-oncological history	**2023**	**Rheumatoid arthritis**
Toxicities grade >= 2 or unknown	**None**	
LVEF	**50%**	
Known allergies	**None**	
Recent surgeries	**None**	

Recent molecular results

Hartwig WGS (22-Feb-2025)

Molecular tissue of origin prediction	**Lung: Non-small cell: LUAD (98%)**
Tumor mutational load / burden	**TML 160 / TMB 14 mut/Mb**
Microsatellite (in)stability	**Stable**
HR status	**Proficient (0)**
Driver mutations	**EGFR C797S, EGFR L858R, KRAS G12C, KRAS G12D**
Amplified genes	**None**
Deleted genes	**TP53 del**
Homozygously disrupted genes	**None**
Gene fusions	**MET(exon13)::MET(exon15) fusion**
Driver virus	**None**
HLA-A	**HLA-A*01:01, HLA-A*02:01**

Trial-relevant IHC results

PD-L1	**Score > 50%**

Standard-of-care options considered potentially eligible

There are no standard of care treatment options for this patient

Phase 2/3+ (or unknown phase) trials in NL that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES	WARNINGS
METC 04 TEDR1 (Phase 2)	Lung cancer C797S cohort	EGFR C797S	NKI-AvL	None
EGFR-C797S-TRIAL (Phase 2)	*EGFR C797S*	*EGFR C797S*	*Elisabeth-TweeSteden Ziekenhuis*	

Trials matched solely on molecular event and tumor type (no clinical data used) are shown in italicized, smaller font.

Phase 1/2 trials in NL that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES	WARNINGS
METC 02 KAYRAS (Phase 1/2)	Dose expansion - monotherapy - NSCLC	KRAS G12D, PD-L1 >= 50.0	Erasmus MC	Variant(s) G12D in KRAS but subclonal likelihood of > 50%
METC 01 IEMOEN (Phase 1)	Dose escalation - monotherapy *(no slots)*	None		Has not exhausted SOC (at least platinum doublet remaining)

International trials that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES
EGFR-BE (Phase 1)	*EGFR L858R*	*EGFR L858R*	*Belgium: Brussels*
KRAS-G12C-TRIAL-DE (Phase 1)	*KRAS G12C*	*KRAS G12C*	*Germany: Stuttgart*

Trials in this table are matched solely on molecular event and tumor type (clinical data excluded).

Molecular Details

Hartwig WGS (EXAMPLE-LUNG-01-T, 22-Feb-2025)

General

PURITY	PLOIDY	TML STATUS	TMB STATUS	MS STABILITY	HR STATUS	DPYD	UGT1A1
50%	2.3	**High (160)**	**High (14)**	Stable	Proficient (0)	*1_HOM (Normal function)	*1_HOM (Normal function)

Predicted tumor origin

	1. LUNG: NON-SMALL CELL: LUAD
Combined prediction score	**98%**
This score is calculated by combining information on:	
(1) SNV types	60%
(2) SNV genomic localisation distribution	70%
(3) Driver genes and passenger characteristics	80%

Other cohorts have a combined prediction of 2% or lower

Key drivers

TYPE	DRIVER	TRIALS (LOCATIONS)	TRIALS IN HARTWIG	BEST EVIDENCE IN EXTERNAL	RESISTANCE IN EXTERNAL
Mutation (gain of function)	EGFR L858R (2/4 copies)		NCT00000007	Approved	
Mutation (gain of function)	EGFR C797S (1/4 copies)	TEDR1 (NKI-AvL)	NCT00000008	Pre-clinical	
Mutation (gain of function)	KRAS G12D (0.3/2 copies)*	KAYRAS (Erasmus MC)			

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Hartwig WGS (EXAMPLE-LUNG-01-T, 22-Feb-2025)

Key drivers

Continued from the previous page

TYPE	DRIVER	TRIALS (LOCATIONS)	TRIALS IN HARTWIG	BEST EVIDENCE IN EXTERNAL	RESISTANCE IN EXTERNAL
Mutation (gain of function)	KRAS G12C (0.3/2 copies)*		NCT00000009		
Deletion	TP53 del				
Known fusion	MET(exon13)::MET(exon15) fusion Domain(s) kept: Tyrosine Kinase Domain(s) lost: Juxtamembrane				

* Variant has > 50% likelihood of being sub-clonal

Other drivers or relevant events

None

Immunology

HLA GENE	TYPE	TUMOR COPY NUMBER	MUTATED IN TUMOR
HLA-A	HLA-A*01:01	1.0	No
	HLA-A*02:01	0.0	No

IHC results

PD-L1	**Score > 50%**

Molecular history

EVENT	DESCRIPTION	2025-02-22 HARTWIG WGS
EGFR L858R (Tier I)	Mutation (gain of function)	VAF 0.5%
EGFR C797S (Tier II)	Mutation (gain of function)	VAF 0.25%
KRAS G12C (Tier III)	Mutation (gain of function)	VAF 0.15%
KRAS G12D (Tier III)	Mutation (gain of function)	VAF 0.15%
MET(exon13)::MET(exon15) fusion (Tier III)	Known fusion Gain of function	Detected
TP53 del (Tier III)	Deletion Unknown protein effect	Detected
TMB		14.0
MSI		Stable

Efficacy evidence

Standard of care options considered potentially eligible

The following standard of care treatment(s) could be an option for this patient. For further details per study see 'SOC literature details' section in extended report.

There are no standard of care treatment options for this patient

There are no standard of care treatment options for this patient

Resistance evidence

There are no standard of care treatment options for this patient

Treatment ranking

EVENT	TREATMENT	SCORE
EGFR L858R EGFR C797S	AFATINIB	2400
EGFR L858R	OSIMERTINIB	1900

On label clinical evidence

EVENT	CKB EVENT	LEVEL A	LEVEL B	LEVEL C	LEVEL D
EGFR C797S	EGFR C797S				AFATINIB *Lung non-small cell carcinoma (2015)*
EGFR L858R	EGFR L858R	OSIMERTINIB *Lung non-small cell carcinoma (2016)* AFATINIB *Lung non-small cell carcinoma (2013)*			

Off label clinical evidence

None

Clinical Details

Clinical summary

Relevant systemic treatment history	**2023-06 to 2025-01**	**Osimertinib**
Relevant other oncological history	**None**	
Previous primary tumor	**None**	
Relevant non-oncological history	**2023**	**Rheumatoid arthritis**
Toxicities grade >= 2 or unknown	**None**	
LVEF	**50%**	
Known allergies	**None**	
Recent surgeries	**None**	

Tumor details

Measurable disease	**Yes**
Lesions	**Liver, Lung**
No lesions present	**CNS, Brain, Bone, Lymph node**

Active medication details

MEDICATION	ADMINISTRATION	START DATE	STOP DATE	DOSAGE	FREQUENCY
St. John's Wort	Oral	01-Feb-2023		300 MILLIGRAMS	1 / 2 DAYS

Trial Matching Details

National trials that are open and potentially eligible (1 trial)

TRIAL	COHORT	MOLECULAR	SITES
EGFR-C797S-TRIAL (Phase 2)	*EGFR C797S*	*EGFR C797S*	*NL: Tilburg, Germany: Stuttgart*

Trials in this table are matched solely on molecular event and tumor type (clinical data excluded).

International trials that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES
EGFR-BE (Phase 1)	*EGFR L858R*	*EGFR L858R*	*Belgium: Brussels*
KRAS-G12C-TRIAL-DE (Phase 1)	*KRAS G12C*	*KRAS G12C*	*Germany: Stuttgart*

Trials in this table are matched solely on molecular event and tumor type (clinical data excluded).

Trials and cohorts that are potentially eligible, but are closed (1 trial)

TRIAL	COHORT	MOLECULAR	SITES	WARNINGS
METC 01 IEMOEN (Phase 1)	Dose expansion - monotherapy	None		Has not exhausted SOC (at least platinum doublet remaining)

Trials and cohorts that are considered ineligible (4 cohorts from 3 trials)

TRIAL	COHORT	MOLECULAR	INELIGIBILITY REASONS
METC 02 KAYRAS (Phase 1/2)	Dose expansion - monotherapy - Colorectum	KRAS G12D, PD-L1 >= 50.0	No colorectal cancer

Trials and cohorts that are considered ineligible (4 cohorts from 3 trials)

TRIAL	COHORT	MOLECULAR	INELIGIBILITY REASONS
METC 03 NO-SEE797ES	Dose escalation - monotherapy	EGFR C797S	C797S in EGFR in canonical transcript
METC 05 PICKME3CA	*Applies to all cohorts below*	None	No PIK3CA activating mutation(s)
	Dose expansion - monotherapy - NSCLC *(closed)*		
	Dose expansion - monotherapy - Other cancer types *(closed)*		Tumor belongs to DOID term(s) lung non-small cell carcinoma

Trials and cohorts that are not evaluable or ignored (0 trials)

None

Potentially eligible open trials & cohorts

METC 01

Potentially eligible	**Yes**
Acronym	**IEMOEN**
Title	**Phase I first-in-human study to evaluate safety of IEMOEN, a new PD-L1 inhibitor in advanced solid tumors**

REFERENCE	EVALUATION
I-03	WARN Has not exhausted SOC (at least platinum doublet remaining)
E-02	UNDETERMINED No measurement found for hemoglobin
E-03	UNDETERMINED No measurement found for absolute neutrophil count

REFERENCE	EVALUATION
E-01	PASS Has no other condition belonging to category autoimmune disease
I-01	PASS Patient is at least 18 years old
I-02	PASS Has solid primary tumor Stage IV is considered metastatic

METC 01 - Dose escalation - monotherapy

Cohort ID	**A**
Potentially eligible?	**Yes**
Open for inclusion?	**Yes**
Has slots available?	**No**

METC 01 - Dose expansion - monotherapy

Cohort ID	**B**
Potentially eligible?	**Yes**
Open for inclusion?	**No**
Has slots available?	**No**

METC 02

Potentially eligible	**Yes**
Acronym	**KAYRAS**
Title	**A phase 1/2 trial for first in-human usage of KAYRAS, a new specific KRAS G12D inhibitor in NSCLC and colorectal cancer**

REFERENCE	EVALUATION
I-04	WARN Variant(s) G12D in KRAS but subclonal likelihood of > 50%
I-03	UNDETERMINED ASAT and ALAT are not present or cannot be evaluated
I-01	PASS Patient is at least 18 years old
I-02	PASS Stage IV is considered metastatic
I-05	PASS PD-L1 expression above minimum of 50.0

METC 02 - Dose expansion - monotherapy - NSCLC

Cohort ID	**A**
Potentially eligible?	**Yes**
Open for inclusion?	**Yes**
Has slots available?	**Yes**

REFERENCE	EVALUATION
I-02	PASS Tumor belongs to DOID term(s) lung non-small cell carcinoma

METC 02 - Dose expansion - monotherapy - Colorectum

Cohort ID	**B**
Potentially eligible?	**No**
Open for inclusion?	**Yes**
Has slots available?	**Yes**

REFERENCE	EVALUATION
I-02	FAIL No colorectal cancer

METC 04

Potentially eligible	**Yes**
Acronym	**TEDR1**
Title	**TEDR1 Trial: A phase II trial to evaluate efficacy of specific EGFR inhibitors in lung cancer**

REFERENCE	EVALUATION
I-1	PASS Patient is at least 18 years old
I-2	PASS Stage IV is considered metastatic Tumor belongs to DOID term(s) lung cancer
I-3	PASS C797S in EGFR in canonical transcript

METC 04 - Lung cancer C797S cohort

Cohort ID	**A**
Potentially eligible?	**Yes**
Open for inclusion?	**Yes**
Has slots available?	**Yes**

Other trials & cohorts

METC 03

Potentially eligible	**No**
Acronym	**NO-SEE797ES**
Title	**Phase I trial for development of NO-SEE797ES, a specific inhibitor for EGFR with C797 mutations but not C797S in solid tumors**

REFERENCE	EVALUATION
I-03	FAIL C797S in EGFR in canonical transcript

METC 03 - Dose escalation - monotherapy

Cohort ID	**A**
Potentially eligible?	**No**
Open for inclusion?	**Yes**
Has slots available?	**Yes**

METC 05

Potentially eligible	**No**
Acronym	**PICKME3CA**
Title	**A phase 1/2 trial of ABC123 +/- platinum doublet in PIK3CA-mutated solid cancer**

REFERENCE	EVALUATION
I-04	FAIL No PIK3CA activating mutation(s)

METC 05 - Dose expansion - monotherapy - NSCLC

Cohort ID	**A**
Potentially eligible?	**No**
Open for inclusion?	**No**
Has slots available?	**Yes**

METC 05 - Dose expansion - monotherapy - Other cancer types

Cohort ID	**B**
Potentially eligible?	**No**
Open for inclusion?	**No**
Has slots available?	**Yes**

REFERENCE	EVALUATION
I-03	FAIL Tumor belongs to DOID term(s) lung non-small cell carcinoma